BrickPlan: A Benchmark for Visual Assembly and Error Recovery Supplementary Material

Anonymous CVPR submission

Paper ID Draft

A. Current Artifact and Evidence Boundaries

This supplement follows the organized dataset and metric specification of 25 October 2026. It supersedes the 5,260-record paper snapshot and its old experiment allocations. Historical model traces and the 96-trial route study remain explicitly version-scoped evidence. Neither supplies formal scores for the current interface.

The artifact's organized catalogue, classification audit, physical replay, source groups, image examples, and scoring functions are bound by hashes. The paper records the current dataset in `evidence/current-dataset.json`, the candidate experiment assignments in `evidence/current-experiment-design.json`, and empty results in `evidence/current-results-template.json`. Older evidence files remain available for provenance, not as current case counts or budgets.

All 5,922 retained execution conditions pass fresh reference replay. This proves witnessed feasibility under the executor. It does not prove that every image is unambiguous, every adaptive event is visible, or that models and humans can solve the tasks. The current release status is program-checked with human image qualification pending. No paid model API is called by the paper build, and no new model or human result is imputed from a reference trajectory.

B. Dataset Accounting and Source Splits

B.1. Records, aliases, and distinct tasks

The organized catalogue has 5,839 retained records: the earlier 5,793 plus 46 complete-original-object conditions. It identifies 119 aliases within a source/configuration/contract: 27 static states equivalent under physical part matching and 92 repeated normal dynamic tasks. Removing their additional task weight leaves 5,720 distinct records. The equivalence test is scoped; two related designs from different source families are not claimed to have undergone global semantic deduplication.

An additional 83 historical M/expanded-adapter aliases are excluded from the global catalogue but preserved as execution conditions. Consequently, 5,839 catalogue conditions plus these 83 adapters yield 5,922 replays. The five implementation banks contribute 3,292 S, 1,712 D, 96 M, 776 expanded, and 46 original-object reference executions. Distinct tasks within these banks number 3,265, 1,620, 96, 693, and 46. Bank labels describe provenance/runtime, not the primary evaluation scopes.

Table S1. **Task types and main candidate counts.** C has no independent bank yet. Controls are scored separately from faults.

Type	Required behavior	Main N
S0	Accept an already-correct stage.	260
S1	Repair a supplied faulty stage.	3,437
D0	Build normally without an external event.	63
D1	Continue through one external event.	1,649
D2	Continue through two or three events.	43
C	Repair from a system-built checkpoint.	Planned

The main set contains 5,452 candidates (3,697 static and 1,755 dynamic): 3,315 local static, 148 whole-object static stages, 234 named-module static stages, 1,534 whole-object dynamic, and 221 named-module dynamic tasks. Another 224 dynamic structural diagnostics and 44 arbitrary-crop dynamic tasks remain outside object main results. Their static counterparts can remain valid local repair tasks; a fragment need not be a recognizable whole object to test static structure correction.

B.2. Source groups and public development material

All 5,720 distinct tasks are assigned by original set or procedural family to 154 source groups, with zero groups crossing splits. Forty-five sources already shown in galleries, examples, or debugging are promoted to public development. Table S2 gives the unified current assignment. This replaces the old S-only 2,286/539/467 split in current paper claims. The historical balanced D panel is also not the current unified test allocation.

Table S2. **Unified source-group assignment.** These are catalogue allocations, not visually admitted or executed model sample sizes. Source groups are disjoint across rows.

Split	Distinct tasks
Train	2,076
Development	400
Public development	2,840
Existing-source test	404
Total	5,720

The existing-source test has 404 distinct tasks, of which 392 are main candidates: 309 local-static tasks (291 S1, 18 S0) and 83 whole-object dynamic tasks (79 D1, four D0). The remaining twelve are historical structure diagnostics. All named-module and complete-original-object examples currently belong to public development. A model comparison may report their performance descriptively, but cannot claim untouched community-source generalization from them.

Source separation does not establish absence from foundation-model pretraining. New-source evaluation needs newly acquired sources chosen before tuning and kept out of public examples. Future community holdouts and longer modules should be reported separately. Split names alone do not certify visual quality or statistical independence of variants.

C. Complete Object and Module Coverage

C.1. Fifty whole-object configurations

The earlier 42 complete-object designs remain, including the chair, table, city car, and crane (Fig. S2). They are intentionally simplified procedural objects. Five longer procedural builds extend coverage: a 21-part bookshelf, 32-part city bus, 25-part bridge, 24-part gantry press, and 24-part cottage. Their larger part counts increase sequence length but do not alone establish difficulty.

Three additional whole objects preserve complete downloaded LDraw source files: the 12-part 4018 ship, 13-part 4017 sea helicopter, and nine-part 4037 helicopter. All original instances, colors, and poses are retained, with zero dropped parts. Wheels or rotors are not removed to make the task easier. They start on an empty white plate. The claim is complete coverage of these source files, not independent confirmation against retail packaging. Together these additions yield 50 whole-object operation configurations, not 50 original community models.

The three original objects contribute 46 conditions: 25 static stages and 21 dynamic tasks. Faults cover missing, extra, color, part, position, and orientation discrepancies where feasible. All normal routes and fault repair witnesses replay legally, and three alternative normal orders are checked. The condition totals must not be mistaken for an equal number of distinct objects.

C.2. Thirteen named remaining modules

Twelve community scenes supply recognizable bodies and explicit remaining modules: cottage roof, racer cabin, seaplane tail, museum roof, truck cab lights, two helicopter upper/tail assemblies, panda head, burger-stand roof/sign, White House roof details, Rockefeller crown, and Brandenburg crest (Fig. S4). These twelve source scenes contain 2,135 instances in the original M inventory, with 99 task placements in its 5–12-part selected modules. The expanded museum roof adds a thirteenth configuration with sixteen active pieces, without adding another source.

The original M adapter retains 96 conditions (four static and four dynamic per module). Expanded conditions introduce additional error families and repeated events. Duplicate controls and cross-adapter aliases do not gain extra weight. Current named-module totals across adapters are 234 static and 221 dynamic distinct tasks. An original M eight-case count is therefore not the current total for that source configuration. The expanded module collection is not limited to three physical errors.

C.3. Static and diagnostic continuity

All 160 source-derived static configurations remain. The original white baseplate migration certified 3,292 raw questions; within-contract physical deduplication now assigns them 3,265 task weights. All 127 earlier static configurations and all 30 earlier dynamic structures are retained in their relevant current main or diagnostic scope. A change in main eligibility does not delete provenance or convert an arbitrary crop into a complete object.

The 26 older non-object dynamic structures remain diagnostic (Fig. S5). Three additional arbitrary community crops, `new-10013-1`, `new-10015-1`, and `new-10016-1`, are also excluded from object dynamic main results. Their local static repair tasks can still be studied in the local structure scope.

D. Error Taxonomy and Structural Descriptors

D.1. Physical labels versus human-inspired causes

E01–E07 denote missing pieces, excess pieces, wrong colors, wrong parts, position/height offsets, non-equivalent orientations, and swapped positional relations. Matching is physical rather than instance-based. A replacement can create both a missing correct placement and an excess wrong placement. A swap means changed positions, not mirrored physical geometry. Direction changes equivalent under supported brick symmetries are not orientation errors.

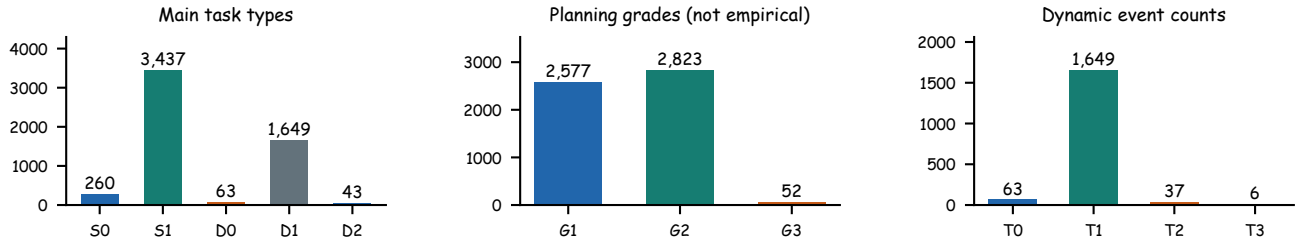

Figure S1. **Current main-candidate composition.** Counts derive from the organized distinct-task catalogue. T0 in the right panel denotes the 63 normal dynamic tasks; static tasks are not in that panel. Planning grades are structural scheduling labels, not measured difficulty.

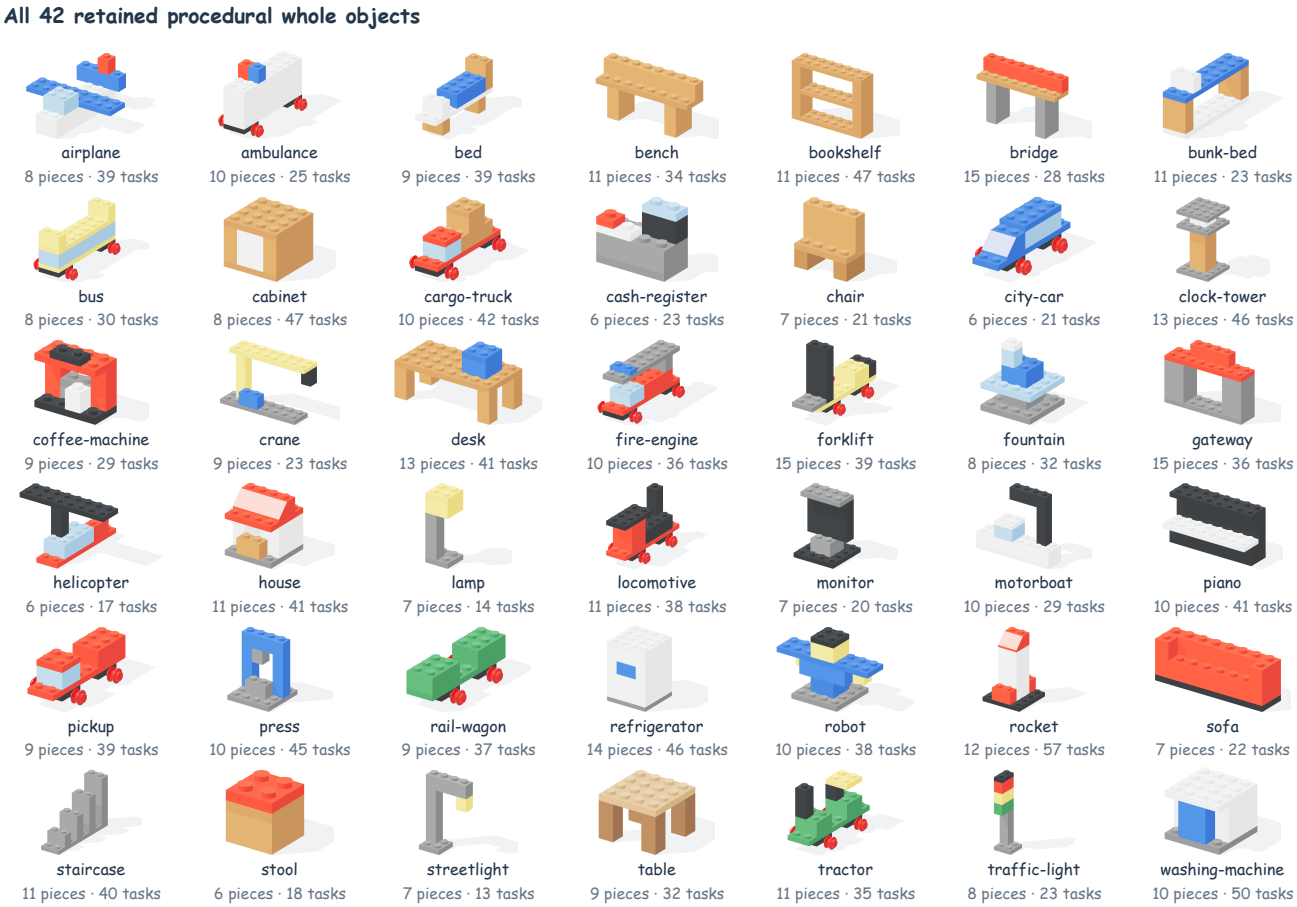

Figure S2. **All 42 retained procedural complete objects.** Each cell gives active parts and distinct current dynamic task count. Compared with historical galleries, repeated controls and excluded human-error scope are removed from the counts. Geometry is unchanged. These simplified objects complement the source-preserving LDraw examples.

P01 marks premature covering that can block a later insertion. In a static stage task, the same cover may simply be premature relative to that stage target; only continued execution tests the access consequence. P02 marks a buried defect with reference dismantling of correct layers. A witness that removes covers does not prove every successful repair must remove the same covers. Repetition omission remains a missing-piece error with a process/cause tag. Simultaneous faults can have several physical labels; they are not repeated temporal events.

T0–T3 record zero to three external interventions. Main D2 includes 43 tasks (37 T2, six T3); the broader catalogue

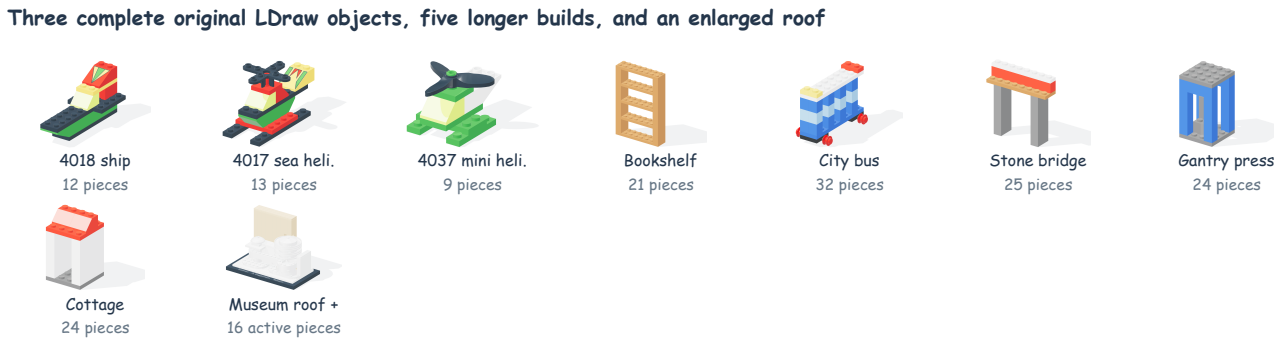

Figure S3. **Nine added operation configurations.** Three complete original LDraw models, five longer procedural objects, and one enlarged museum roof. Counts are distinct dynamic tasks; static siblings are additional tasks. The roof retains the full building context and manipulates only its named remaining portion. All scenes are rendered natively at 3200×2400.

Figure S4. **All thirteen named operation configurations.** Full source scenes establish semantic context; active-part counts refer only to the remaining modules. Labels count active operation pieces, excluding supplied bodies. The enlarged roof is a second configuration on the existing museum source.

has 47 repeated-event tasks, including four archived-crop diagnostics. These populations must not share an unexplained denominator.

Human-inspired tags H01–H11 and H13 are twelve approved categories motivated by community reports. H12 and H14–H17 are excluded from the current task selection; H18 remains deferred. The inventory is not a live remaining-piece count, so no leftover-count reasoning claim is made. Synthetic category balance is not an estimate of real human mistake frequency, and online reports are not independently collected participant trajectories.

D.2. Migration labels and reference burdens

The old static migration retained 3,025 source-linked records and replaced 267 incompatible cases. Its raw ledger contains 2,144 direct replays, 834 dismantle/rebuild witnesses, 37 held-component witnesses, ten corrected board-penetrating starts, and 267 substitutions. This historical ledger sums to 3,292 raw conditions and precedes the new deduplication. Replacement-state labels are resolved using qualified candidate kinds after physical multiset matching; they are not a new error class. Height-layer cases map to actual displaced poses. No labelled static error in the organized catalogue is physically equivalent to its target.

Build descriptors count the current target, not unused future pieces. B1 is at most twelve active pieces, B2 is thirteen to twenty-four, and B3 is at least twenty-five. R0 is a control. R1 has at most two reference repair moves and no reference removal of correct covers. R3 has such a removal or at least seven reference moves. R2 covers remaining ordinary rework, including three to six moves and relevant multiple-defect cases. These rules describe the available witness; optimal repair search has not been proved.

The provisional grade is G3 if B3, T3, or both T2-or-more and R3. Otherwise it is G2 if B2, R2/R3, or T2; the remaining tasks are G1. Main counts are 2,577 G1, 2,823 G2, and 52 G3. Empirical model/human results must validate these descriptors before calling them easy, medium, and hard. Visibility remains an independent admission field; invisibility is not merely a higher difficulty grade.

Figure S5. **All 26 historical dynamic structural configurations.** The 224 distinct tasks test execution mechanisms and local repair. They are not included among the 63 object/module dynamic configurations or in object-construction main scores.

Table S3. **Main-candidate multilabel coverage.** A task can occur in several rows. Process tags are distinct from the seven physical labels. The absent module-swap stratum is a coverage gap. Counts include normal controls only when a label genuinely applies (normally none).

Label	Meaning	Static	Whole dynamic	Module dynamic
E01	Missing piece	1,361	209	60
E02	Extra piece / local added layer	356	167	36
E03	Wrong color	1,458	426	60
E04	Wrong type, size, or thickness	339	185	15
E05	Position or height offset	804	227	32
E06	Non-equivalent orientation	489	130	25
E07	Positional relation / swap	31	20	0
P01	Premature cover / access obstruction	17	23	4
P02	Buried repair	36	219	22

E. Exact Input and Action Interface

E.1. Public input and private evaluation state

The current contract is `brick-atlas-direct-actions-full-context/1`. The common local runner uses existing physical engines and emits one direct action schema. S and D no longer require a different outer `action` object. Historical interface results retain their old input version. The public gallery is an author-facing display of labels and witnesses, not a blind model input. Only the observation adapter is used for a model attempt.

Each matched current/target scene is 800×600 pixels on a 1600×636 comparison canvas. Native partial-manual pages can be 1600×1040. Isometric, rear, and top views are available through supported inspection. Named modules retain full-object context in both current and target images. A publication crop does not redefine the input framing. Actual image count, dimensions, hashes, and provider processing must be logged per request; the number of images is not a constant cost inferred from the number of scene roles.

The model receives neutral rules and anonymous action handles alongside RGB. This is not pure image-only input. It does not receive raw pose maps, error codes, family labels, reference correction actions, task type, planned accidents, or hidden event times. The first decision identifies that no prior action exists. Public names and image paths must not encode the correct candidate. Observations are checked for prohibited fields.

E.2. Direct actions and finite candidates

A response is one JSON object. Examples are:

Table S4. **Roles of current input information.** The acceptance target is unique. Files and paper layouts do not define image roles implicitly. Information varies by track; unsupported queries are not silently provided.

Role	Content	Scoring interpretation
Acceptance target	Required static stage or completed dynamic object/module	Sole endpoint for this task.
Current scene	Actual state after own action and any hidden event	Authoritative world observation.
Previous scene	State before the preceding own action	Temporal evidence; absent at the first decision.
Supplied context	Whole body around a named remaining module	Visible fixture; excluded from movable-part scores.
Reference manual	Isolated part and state after one reference placement	Optional route, where supported; future pages do not change a static stage goal.
Inventory	Part cards, colors, and anonymous handles, with spares	Catalogue, not a live count of remaining pieces.
Candidate preview	Hypothetical selected part/pose overlay	Query aid, not a correctness label or executed action.
Legality feedback	Physical result or reason for rejection	Separate from image evidence and semantic correctness.

```
{"type":"attach","partId":"A023",
  "candidate":"C00523"}
{"type":"detach","partId":"A023"}
{"type":"inspect","view":"top"}
{"type":"finish"}
```

These handles illustrate an interface, not universal answers. Attach and detach each manipulate one physical part. Inspection can request a supported view, part, preview, inventory page, or manual step. It does not change geometry or itself trigger an accident. Multiple-action scripts, hidden state replacement, and unsupported field combinations are invalid in the default interface. Human participants operate buttons; they do not type these objects or write explanatory text.

The main action space selects anonymous candidate poses. It is not unrestricted six-degree-of-freedom prediction. Free coordinates require another adapter and output-localization experiment. Likewise, text-state and RGB-plus-state conditions are planned separately. Explicit state can reveal occluded geometry, so such a comparison measures added state information rather than equal-information modality alone.

E.3. Physical checks and budgets

Attachment checks supported pose/connection, endpoint collision, and a straight insertion path. Detachment checks extraction. A rejection changes no geometry; a later correct finish remains successful. Feedback and candidate previews can aid planning and must be held constant or ablated explicitly. Plain rectangular brick symmetries are supported only where justified. Decorated and asymmetric pieces are not assumed rotationally invariant. Terminal comparison fixes the construction frame rather than globally aligning arbitrary assemblies.

The proposed comparison cap is $\max(40, 6N + 20)$ model decisions, including inspections and malformed outputs. Applicable native caps, such as the original module manipulation limit $4N + 12$, remain recorded. Native limits are not claimed identical across every engine. A formal comparison must declare the effective caps per track before running and use the same caps for every model on the same task. A reference route is feasible, not necessarily shortest.

F. Static Metrics in Detail

Let I, F, G denote initial, final, and target multisets of canonical shape/color/symmetry-aware poses for movable pieces only. Immutable context is excluded. The endpoint success indicator requires explicit `finish` and $F = G$. A rejected physical request is not accepted as a state transition; its existence does not invalidate a subsequent successful completion. Model budget exhaustion is an unsuccessful attempt, not an implicit finish at a correct intermediate snapshot.

$$\mathrm{TP} = |F \cap G|, \quad \mathrm{FP} = |F - G|, \quad \mathrm{FN} = |G - F|, \tag{1}$$

$$\mathrm{F1}_{\mathrm{place}} = \frac{2\mathrm{TP}}{2\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}, \tag{2}$$

$$E_0 = |G - I| + |I - G|, \quad E_T = |G - F| + |F - G|, \tag{3}$$

$$\mathrm{ER} = 1 - E_T / E_0, \tag{4}$$

$$H = I \cap G, \qquad \mathrm{DR} = |H - F|/|H|. \tag{5}$$

Undefined denominators are N/A. F1 of two empty multisets is defined as one. ER can be negative. If a wrong-color brick is removed but its correct-color counterpart is not installed, residual falls from two to one: ER is 50%, but exact SR is zero. This does not mean that one of two human mistakes was fixed or that a globally shortest repair was used.

For S0 controls, FR records any accepted attach/detach during an initially correct task. A remove-and-restore sequence has FR one and can still finish correctly. DR instead records final damage only; temporarily removing correct work to access a buried error does not incur DR if re-

stored. Normal acceptance, unnecessary manipulation, and fault repair therefore need separate denominators.

Original missing recovery is $|(F-H)\cap(G-I)|/|G-I|$. Original excess clearance is $1-|(F-G)\cap(I-G)|/|I-G|$. Moving an excess brick to a different wrong pose can clear its original excess signature while still failing the task; these are diagnostics, not replacements for SR/F1. Type-plus-color, shape-plus-pose, and color-plus-pose F1 expose attribute combinations, not independent visual recognition accuracies. A uniform connection-adjacency F1 is planned but not implemented in the common scorer.

G. Dynamic Metrics and Pair Eligibility

G.1. Normal building and raw intervention outcomes

BCR uses valid normal D0 runs. ICR uses valid D1/D2 runs assigned an intervention, whether or not the event was reached. TR is actual events over planned events, with per-task exposure counts. An untriggered correct completion is raw completion, not successful repair. A partial one-of-two exposure remains visible in the ledger and does not enter all-event recovery. An empty recovery denominator is N/A.

C denotes future system-built checkpoint continuation. Its CSR measures repair plus remaining completion from that checkpoint. The prefix must be described as supplied by the system, not assembled by the model. There is no independent C bank in this snapshot, and CSR cannot replace D1/D2 continuous-construction evidence.

G.2. Shared-prefix construction and incident effect

Execute the common prefix once. Save environment state, conversation, images, handle mapping, used budget, and random settings. Clone these into control and intervention continuations at the admissible incident opportunity. Keep targets, inventory, action vocabulary, budgets, and model revision fixed. After branching, allow each continuation its own legal order. Two separate normal/fault runs with the same nominal seed are not verified shared-prefix pairs.

The eligible population Q requires valid branches, verified shared history, all planned events actually applied, and verified incident effects. An event that removes an already-wrong brick or helps an incorrect agent cannot automatically support a standard repair claim. Record pre-existing errors and the realized edit. A correct future cover can block assembly without adding a target residual; verify access obstruction through the executor rather than demand a residual increase. Adaptive event locations can have trajectory-dependent difficulty.

For $c_i, f_i \in \{0,1\}$ on Q, CRR is $\sum_i c_i f_i / \sum_i c_i$. Its denominator excludes failed normal branches. The signed paired difference is $\Delta\mathrm{SR} = |Q|^{-1}\sum_i (c_i - f_i)$ and retains all four outcome cells, including fault-only success. It can be negative. Report planned pairs, technically valid branches, normal failures, untriggered pairs, partial exposure, unverified effects, and eligible counts. A high conditional rate with poor normal-build or exposure coverage must not be presented alone.

G.3. Multiple events and recovery while progressing

For a realized event, let B be the pre-event movable multiset and A the post-event multiset. Relative to full target G, obligations are

$$L = (B \cap G) - A, \qquad X = (A - G) - (B - G). \quad (6)$$

At a later state, the lost correct quantities must be restored and the incremental excess quantities removed. There is no requirement to return exactly to B: legitimate new target placements are permitted. Pre-existing agent defects remain separate. Windows end immediately before the next external intervention so a later accident is not counted as regression from the earlier repair.

Count all decisions, including inspections, invalid outputs, and physical rejections, toward first-recovery latency. Report latency conditional on observed recovery together with unrecovered proportion. Also report window-end recovery and retention over at least two follow-up decisions. An observed recurrence is a retention failure; a correct short window is N/A with its sample count. An early-cover process error with no L, X positions must not receive automatic zero latency. Use completion and verified process measures instead. Repair@2/@4/@8 is reserved for sufficiently observed single-event or checkpoint windows; next-event and technical truncation are listed separately.

All-event D2 CRR requires successful normal control, all planned events, verified effects, and terminal success. Per-event repair reports use actual effective events and their windows. The two measures must not be confused when a first repair succeeds but the second event is never reached. The common metric functions implement eligibility and window logic; formal batch-runner logging still needs integration.

G.4. Detection is a separate task

The continuous agent is not required to state whether it noticed an anomaly. Failure can arise from perception, action selection, ordering, or premature stopping. Successful recovery does not prove conscious or explicit detection either. Independent detection uses fixed normal and fault snapshots with balanced change/no-change coverage, and reports precision, recall, F1, false-positive rate, and multi-label location-set F1. It remains a planned task bank and must not interrupt the primary assembly episode or manufacture detection labels from GT actions.

H. Technical Status, Aggregation, and Cost

H.1. Ability failures and censored runs

`completed` and `model_budget_exhausted` are ability-valid statuses; the latter is a failure. A premature finish with a wrong target is also failure. Final `service_error`, `harness_error`, and `external_stop` do not receive ability scores. A transient service error recovered at the same decision can lead to a valid completed run; its charge and retry are retained. It is incorrect to censor every completed attempt merely because one retry occurred.

Parsing failure, parsed-but-invalid schema/handles, and physical rejection are separate events. Accepted attach/detach over well-formed manipulation attempts defines legal-action rate. Every attempted request and decision is retained in cost and failure reporting. Neither a 403/429/timeout nor JSON-format failure should be relabeled as a visual error.

Report valid-run coverage and completed/planned-attempt operational rate alongside ability SR. The latter planned denominator combines access and task ability and must be named accordingly. All compared models receive the same predeclared task list; service failures do not justify substituting other tasks. A common-valid-subset analysis can be added with its scope and N explicitly recorded.

H.2. Aliases, macro aggregation, and uncertainty

Group every result by model, input version, track, and task type. Normalize canonical-task weights so duplicate control conditions and interchangeable-ID records do not inflate performance. Average within object and then across objects; additionally report source-macro results by original set or design family. Task-micro rates are references. Static S0 and S1, and dynamic normal/intervention tasks, remain separate. Multilabel error rows cannot be summed as independent task populations.

Source-cluster bootstrap intervals are planned, with 2,000 group resamples retaining variants and paired branches together. Sparse cells remain descriptive. One attempt per model/condition does not establish run-to-run stability. The common scorer implements grouping and macro weights; source-clustered intervals have not yet been integrated into formal model reporting.

H.3. Requests, actions, and financial reporting

Record accepted moves, all manipulation attempts, inspections, decisions, requests, actual image count/dimensions, input/output tokens, wall time, and billed dollars. Divide manipulation attempts by a feasible witness only when that witness is nonzero. The ratio can be below one and is not SPL or globally optimal efficiency. Report zero-witness S0 via FR.

Count shared prefixes once, both continuations, and billed retries or failed calls. USD per successful task is N/A when none succeeds. A service retry repeats the same request without silently restarting the episode or selecting a better answer. The old 8,030-work-unit plan and old budget estimates apply only to the historical snapshot. Current financial estimates require a small current-interface pilot per selected model and a declared admitted case list; no current dollar total is asserted here.

I. Reference and Historical Diagnostics

I.1. Replay and visibility admission

The current replay audit has 5,922 passes and zero failures, including aliases and controls. Source and reference hashes are retained. The three complete LDraw objects have 36 faulty reference conditions and ten controls. Every checked faulty condition changes at least 128 pixels by at least 24 RGB levels in one of three 800×600 views. These reference-image thresholds are machine screening, not human perception guarantees.

Old M and expanded-module image checks used cropped framing. They do not certify current full-context readability, and new adaptive locations need their own observation-level check. Legacy low-contrast and latest-action-cancellation flags are retained separately. A removed newly installed part can make previous and current scenes equal, requiring action-outcome monitoring rather than ordinary change detection. Current-interface human qualification remains pending for all formal perception claims.

I.2. Historical alternative-order study

The original twelve-module study uses two routes and four conditions per route: control, missing piece, wrong color, and extra plate. Its 96 route–condition trials are not the current 5,720-task dataset. Alternative routes were found by legal swaps and preferentially selected to differ at the trigger threshold. Nine modules have a different physical threshold state; three have a different order but the same threshold state. This selection demonstrates regression cases, not the distribution of a learned agent's orders.

Exact-reference triggers have opportunity on 36/36 reference-route and 9/36 alternative-route faults. Actual-state selectors trigger 36/36 in both groups. Every normal route reaches the correct target, while a comparator requiring every manual prefix accepts zero of twelve alternatives. Blind continuation finishes 24/24 controls and 0/72 faults; a privileged inverse-and-continue policy finishes 96/96. Privileged state access makes this a reachability check, not a visual baseline. The recorded source hashes and route indices support reproduction without a model API.

Table S5. **Historical selected alternative routes.** Integers identify reference placements, not compulsory times. This table is unchanged protocol evidence from the original twelve modules.

Object	Alternative reference-position order	Different threshold state
Cottage	1, 2, 4, 3, 5, 6	Yes
Racer	1, 6, 3, 4, 5, 2, 7, 8, 9, 10, 11	Yes
Seaplane	1, 6, 3, 4, 5, 2, 7	Yes
Museum	1, 2, 3, 5, 4, 6, 7, 8, 9, 10, 11, 12	Yes
Repair truck	1, 2, 3, 5, 4	Yes
Orange helicopter	1, 3, 2, 4, 5, 6, 7	No
Police helicopter	1, 2, 3, 5, 4, 6, 7, 8, 9, 10	Yes
Panda	1, 2, 3, 5, 4, 6	Yes
Burger stand	2, 1, 3, 4, 5, 6, 7	No
White House	1, 2, 3, 5, 4, 6, 7, 8, 9, 10	Yes
Rockefeller Center	1, 2, 3, 5, 4, 6, 7, 8	No
Brandenburg Gate	1, 2, 3, 4, 5, 7, 6, 8, 9, 10	Yes

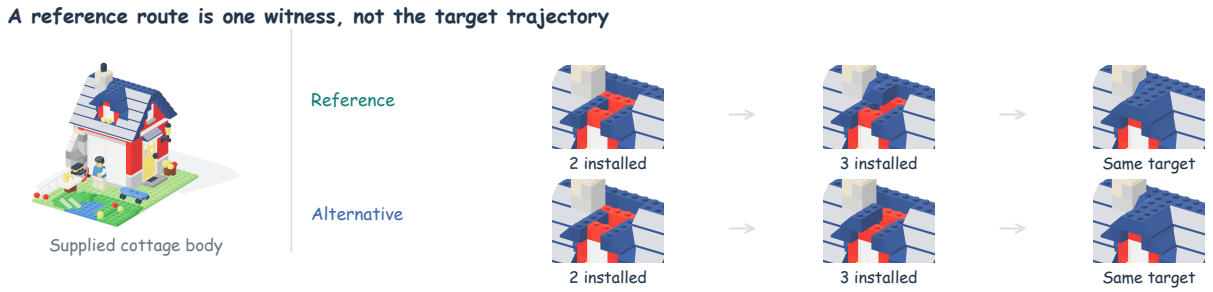

Figure S6. **One target, different legal intermediate states.** The supplied cottage anchors two views of its roof. Both routes share their first two placements; their third placements differ, yet both reach the same six-part target. The two prefixes and endpoints are legally replayed. Publication crops share a viewport across both routes and do not redefine model observations.

I.3. Archived car model and human checks

The historical six-part-car pilot used the provider route `google/gemini-4-argon`, seed 41, and a 40-decision cap. Three older-interface episodes finished: control (18 decisions, eight legal moves, recorded USD 0.1950043), extra layer (14, seven, USD 0.2022685), and removed piece (14, seven, USD 0.2140114). These charges are archived observations, not current prices. The manual, bank, and interface differ from the current contract, so none populates current result tables.

The main filmstrip shows one GT witness and two of those model traces. All 32 selected traces' decisions reproduce saved state hashes and feedback. The extra-layer run uses another legal order. The normal run legally places a piece at the wrong pose, then removes and restores it. Its red frame is a temporary mistake, not terminal failure. Separate normal and fault runs diverged before the event and are not causal shared-prefix pairs. Inspections and some intervening placements are omitted in the eight-column display and disclosed in its caption.

One earlier participant independently submitted eleven correct binary change judgments; a twelfth exposed-answer item was excluded. Two more reviewers were reported to agree without independent submission records. This is neither a three-rater study nor timed assembly performance, and does not qualify all current images. Earlier interface-development 403 errors and no-progress traces likewise do not supply current population-level ability conclusions.

J. Planned Experiments and Empty Result Tables

All nineteen reserved result tables E1–E19 contain unmeasured dashes. Their machine-readable performance fields are null; known assigned case counts are metadata, not scores. All nineteen sheets are grouped in the result-panel appendix; the main paper presents the study questions and candidate roster. The current experiment manifest records candidate IDs and audited existing-source allocations, not a completed admission procedure or implemented paired API runner. Independent C, detection, text-state, and coordinate banks still require construction/integration.

Table S6. **Experimental design references.** These motivate analyses, not direct comparisons of scores across incompatible tasks or input spaces.

Work	Relevant evaluation element	Adaptation here
LEGO-Puzzles; PhyBlock [3, 4]	Task breakdown, human comparison, complexity	Scope-specific main tables, structural strata, common operation panel.
SCANet [7]	Correction and new misplacement	Separate ER, DR, original-defect recovery, and S0 FR.
Break and Make [6]	Structure reconstruction and partial quality	Canonical movable-placement F1 with fixed construction frame.
AsgardBench [5]	Observation and feedback controls	Manual/history/view/feedback and input-state conditions.
RoboRecover [2]	Recovery from deviations and cross-policy analysis	Qualified shared-prefix effect, checkpoint and same-state recovery.
BrickBench [1]	Validity, quality, human judgments, and cost	Distinct legality, goal, human, and efficiency reports.

Table S7. **Named evaluation candidates retained from the earlier plan.** Requested IDs below are planning entries, not verified current availability, account access, or pinned served revisions. Recheck image support and record the actual API revision or checkpoint commit, engine, precision, image processing, and decoding before execution. All rows remain unmeasured.

Model	Access	Requested API ID / checkpoint
GPT-6.1 Sol	API	`gpt-6.1-sol`
Claude Sonnet 5.5	API	`claude-sonnet-5-5`
Gemini 3.8 Flash	API	`gemini-3.8-flash`
Claude Haiku 5.5	API	`claude-haiku-5-5`
Qwen3.5-9B	Open	`Qwen/Qwen3.5-9B`
Qwen3.5-27B	Open	`Qwen/Qwen3.5-27B`
InternVL3.5-8B	Open	`OpenGVLab/InternVL3_5-8B-HF`
Qwen2.5-VL-7B	Open	`Qwen/Qwen2.5-VL-7B-Instruct`
Qwen2.5-VL-32B	Open	`Qwen/Qwen2.5-VL-32B-Instruct`
Gemma 3-12B	Open	`google/gemma-3-12b-it`

J.1. Named model candidates and admission

The roster retains four previously selected API candidates and six open visual candidates, including older Qwen2.5-VL and Gemma 3 baselines. No claim is made that these names are the newest available or all account-accessible. Requested names are historical planning entries; verify vision support, requested ID, actual served revision or checkpoint commit, inference engine, precision, image processing, decoding, output budget, and context before any formal run. An unavailable candidate remains not run rather than being silently replaced.

Use one run per model/condition; seed 41 controls handle mapping and provider randomness only where supported. Four API versus six open models does not isolate architecture, training, or size effects. No frontier-model coverage is claimed. Use Sol, Flash, and Qwen 27B as the preselected diagnostic subset. Scoring summaries always bind the input version, exact model revision, track, and task type.

Formal evaluation uses a common predeclared task list after visual and interface admission. Existing-source test results are separate from public-development module/object results. The candidate list covers all main configurations, but its size is not the final admitted N. Do not use the former 467-question static or 168-fault dynamic panel as if it were the new unified test allocation. Keep all rejected candidates and reasons in an admission ledger, and create untouched community sources before making new-source claims.

J.2. Correction, error, and structural analyses

E5 and E6 distinguish the seven physical error labels with current assignment counts. Repeat process-tagged and simultaneous-error analyses separately. E7 reports part-level restoration and collateral damage by static scope. E8 stratifies B/R/T and provisional G; include sample counts and track/task type in every estimate. Richer labels do not justify pooling normal controls with repair cases or treating a multilabel count as independent examples.

J.3. Repeated events and checkpoint continuation

E9 distinguishes two-event from three-event conditions. Save each event's pre/post state, own-action state before the

edit, realized event effect, and window boundary. Whole-episode success, per-event repair, and retention have different denominators. Failed controls, partial exposure, and short windows remain visible. Repair after the second event must not be attributed to the first without its proper window.

E10 is a future checkpoint study. Its system-built prefix separates ability to construct the first half from ability to repair a supplied midway state. Use identical checkpoint states for every model and list which earlier work is supplied. Do not report it as continuous D1/D2 performance or fill a table before the bank and logging exist.

J.4. Input and output conditions

E4 changes one supported input factor at a time: prior scene, manual, candidate preview, or legality feedback. Recompute shared prefixes if input changes alter construction. Manual removal is unavailable for legacy local-static tasks without a manual. A one-view ablation must also remove alternative inspection views. Provider resizing is a measured property, not assumed equal just because native scenes match.

E11 distinguishes default RGB with neutral text from explicit text state and RGB-plus-state. Extra state can expose hidden pieces; label its privileged scope. A text-state adapter must specify current placements, inventory, pose convention, and inspection equivalents without leaking fault labels or target-correct actions. E12 compares candidate IDs to free position/orientation output only after building a matched coordinate adapter. These are different burdens, not automatically a fair single leaderboard.

J.5. Detection, cross-model repair, and robustness

E13 uses independent fixed snapshots with controls and faulty scenes. Ask for anomaly/no anomaly and a location set; humans use buttons. Use multilabel type/location scoring rather than forcing multiple errors into one choice. Report readability and interface failures alongside valid precision/recall/F1. This probe is separate from continuous assembly and does not establish whether a particular failed agent noticed a fault. Its bank and adapter are planned.

E14 saves the first accepted off-target placement from a normal source-model run, then seeks a legal recovery witness. Select states without observing recoverer success; retain no-state/no-witness exclusions. Each recoverer receives the identical state, inventory, target, and fresh context with source identity hidden. Diagonal restart performance differs from uninterrupted self-repair. This is a proposed selected-state analysis, not an existing bank or an all-failure distribution.

E15 fixes geometry and camera while varying delivered resolution, current-only input, or available views. Record actual images/tokens and visibility under each condition. E16 compares the closed loop with an open-loop action script, explicit checking prompt, or larger decision allowance. An open-loop script still executes through one-brick physical checks. Keep manipulation caps fixed when doubling decisions, record native supported reasoning settings, and never score private reasoning. Reuse identical baseline attempts rather than charging or counting them twice.

J.6. Human operation, efficiency, and failure ledger

The proposed common panel selects twelve normal/fault pairs in each of three scopes: static repair, whole-object dynamic, and named-module dynamic. Its 36 pairs give 72 conditions. Six independent participants operate 36 conditions each; each condition gets three submissions and nobody sees both pair members. This is a plan for 216 operations, not a completed measurement. Independent participants do not form cloned-history causal pairs. Continuous human trials report BCR, ICR, and exposure; a system-supplied common prefix is a checkpoint task, labeled C. Human CRR requires an independently justified pairing design rather than copying agent eligibility assumptions. Current-source case selection and new schedules must be fixed after qualification; the old schedule is historical.

Humans see the same task information and use graphical actions without prose or JSON. Train on separate practice scenes, balance normal/fault assignments, and record completion, exposure, action count, time, and optional unreadable/unusable buttons. Cluster human analyses by both source and participant. Report actual independent submissions; verbal agreement cannot be copied into additional participant rows.

E17 captures image/request/token/time/money and action effort for each model and condition. E18 holds human and common-model outcomes. E19 retains service, schema, physics, normal-build, exposure, and budget status. Read each rate with its valid/admitted/planned counts. A pilot for the current input contract is needed before estimating the price of full evaluation. No new API cost estimate or measured model rank is asserted by these empty tables.

K. Rendering, Attribution, and Remaining Limits

Publication scenes are rendered from the certified LDraw triangle geometry at 3200×2400 pixels using physically shaded materials, studio lighting, multisample antialiasing, and contact shadows. This presentation pipeline changes neither the task state nor the evaluation RGB. Current/target pairs and trajectory groups share camera, lighting, and publication viewport; canvas whitespace is trimmed without distorting geometry. Composite exports use 600 dpi, with vector labels, arrows, and charts. The eight-state filmstrips retain original state/action provenance. Paper-only coral callouts and correctness rules never enter blind model input.

Native interface scenes retain their declared 800×600 resolution and full context. The manifests record scene states, asset hashes, viewports, renderer policy, and figure exports. Public image admission must use the native interface, not these higher-resolution publication illustrations.

Original LDraw files retain authors, set IDs, URLs, and license headers. Source-derived assets remain subject to their attribution and library terms; community authorship is not reassigned. LEGO is a trademark of its owner, with no claimed affiliation. A future public dataset card must preserve source licenses and known reuse constraints.

Physical legality refers to the rigid geometric executor. Straight insertion/extraction, finite candidates, fixed context, and conservative bounds for unsupported surrounding meshes simplify real manipulation. Articulated states are rigid poses; hinge motion, clutch forces, flexibility, stability, slipping, and grippers are outside scope. No real-robot performance follows from these replays. Only three whole objects are complete original LDraw files, long community modules are few, G3 is sparse, and untouched new sources remain to be acquired.

Before empirical claims, finish current-interface image/human admission, pair-prefix/event-effect logging, repeated-event windows, API cost collection, and source-clustered intervals. Build the independent C and detection panels and alternative input/output adapters before filling their tables. Program checks, metric tests, and a finished paper layout are not evidence of CVPR acceptance or completed model evaluation.

L. Reserved Detailed Result Panels

These are E1–E19, grouped as result sheets rather than filling the main text with unmeasured grids. Dashes remain unmeasured. Assigned counts describe the candidate catalogue and do not substitute for valid, exposed, or qualified paired sample counts.

Table S8. **E1: static repair (unmeasured).** Local-static panel; repeat the same schema separately for whole-object and named-module stages. SR/F1/ER/DR use S1; FR uses S0. Source- and object-macro estimates, valid counts, and coverage accompany results. Dashes are unmeasured, never zero.

Model	SR ↑	Place F1 ↑	ER ↑	DR ↓	FR ↓	Valid N
GPT-6.1 Sol	—	—	—	—	—	—
Claude Sonnet 5.5	—	—	—	—	—	—
Gemini 3.8 Flash	—	—	—	—	—	—
Claude Haiku 5.5	—	—	—	—	—	—
Qwen3.5-9B	—	—	—	—	—	—
Qwen3.5-27B	—	—	—	—	—	—
InternVL3.5-8B	—	—	—	—	—	—
Qwen2.5-VL-7B	—	—	—	—	—	—
Qwen2.5-VL-32B	—	—	—	—	—	—
Gemma 3-12B	—	—	—	—	—	—

Table S9. **E2: dynamic construction and recovery (unmeasured).** W/M: whole-object/named-module scope. BCR uses D0; CRR requires normal success and verified complete exposure; Δ uses eligible pairs without filtering control success. Report D1 and D2 separately. ICR, TR and eligibility counts are retained in E3/E6/E9 and the ledger.

Model	W BCR	W CRR	W Δ	M BCR	M CRR	M Δ
GPT-6.1 Sol	—	—	—	—	—	—
Claude Sonnet 5.5	—	—	—	—	—	—
Gemini 3.8 Flash	—	—	—	—	—	—
Claude Haiku 5.5	—	—	—	—	—	—
Qwen3.5-9B	—	—	—	—	—	—
Qwen3.5-27B	—	—	—	—	—	—
InternVL3.5-8B	—	—	—	—	—	—
Qwen2.5-VL-7B	—	—	—	—	—	—
Qwen2.5-VL-32B	—	—	—	—	—	—
Gemma 3-12B	—	—	—	—	—	—

Table S10. **E3: paired outcomes (unmeasured).** Repeat for each model, dynamic scope, and D1/D2. n_{cf} counts control/fault success. CRR uses $n_{11}/(n_{11}+n_{10})$; signed Δ uses all four cells. Only prefix-, exposure-, and effect-qualified pairs enter.

Scope	n_{11}	n_{10}	n_{01}	n_{00}
Whole / D1	—	—	—	—
Module / D1	—	—	—	—
Whole / D2	—	—	—	—
Module / D2	—	—	—	—

Table S11. **E4: input controls (unmeasured).** Sol, Flash, and Qwen 27B are preselected diagnostic candidates. Report track-specific SR/BCR/CRR and counts. Manual removal applies only where a manual exists. Explicit state is privileged.

Condition	Sol	Flash	Qwen
Default RGB interface	—	—	—
No previous scene	—	—	—
No manual (supported tasks)	—	—	—
No candidate preview	—	—	—
Reduced legality feedback	—	—	—
Explicit state assistance	—	—	—

Table S12. **E5: static error breakdown (unmeasured).** Assigned counts are current main-static multilabel coverage, including process-tagged cases. Repeat results by source scope and model; controls are separate.

Label	Assigned	SR	F1	ER	DR
E01 Missing	1361	—	—	—	—
E02 Extra	356	—	—	—	—
E03 Color	1458	—	—	—	—
E04 Part	339	—	—	—	—
E05 Position	804	—	—	—	—
E06 Orientation	489	—	—	—	—
E07 Swap	31	—	—	—	—

Table S13. **E6: dynamic errors (unmeasured).** W/M assignment counts are multilabel. The module-swap cell has zero assigned tasks and yields N/A, not a model failure. ICR uses valid assigned runs; CRR and Δ use their declared paired populations.

Label	W assigned	M assigned	ICR	TR	CRR	Δ
E01 Missing	209	60	—	—	—	—
E02 Extra	167	36	—	—	—	—
E03 Color	426	60	—	—	—	—
E04 Part	185	15	—	—	—	—
E05 Position	227	32	—	—	—	—
E06 Orientation	130	25	—	—	—	—
E07 Swap	20	0	—	—	—	—

Table S14. **E7: stage-repair quality (unmeasured).** Repeat for each named model and each scope. Missing/excess rates track only original defects; they cannot replace exact SR. Context is excluded. S0 false repair has a separate denominator.

Scope	SR	Place F1	Missing recovered	Excess cleared	DR	S0 FR
Local structure	—	—	—	—	—	—
Whole-object stage	—	—	—	—	—	—
Named-module stage	—	—	—	—	—	—

Table S15. **E8: structural strata (unmeasured).** Counts include all main candidates. Static B uses the stage target; R uses a feasible witness. Report outcomes by track/task type rather than pooling these strata into one score. G is provisional, not measured difficulty.

Stratum	Assigned	SR/BCR	CRR	Eligible N
B1	4873	—	—	—
B2	531	—	—	—
B3	48	—	—	—
R0	323	—	—	—
R1	2656	—	—	—
R2	411	—	—	—
R3	2062	—	—	—
T0	3760	—	—	—
T1	1649	—	—	—
T2	37	—	—	—
T3	6	—	—	—
G1	2577	—	—	—
G2	2823	—	—	—
G3	52	—	—	—

Table S16. **E9: repeated interventions (unmeasured).** Main D2 has 37 two-event and six three-event tasks. All-event CRR requires successful normal control and complete effective exposure. Partial exposure is retained in the ledger. Retention needs sufficient follow-up.

Events	Assigned	All-event CRR	Event recovery	Retention	Partial N
2	37	—	—	—	—
3	6	—	—	—	—

Table S17. **E10: checkpoint continuation (planned bank).** The system builds the prefix; this is not agent-built progress. No independent C bank or current CSR results exist. Repair@k requires adequate observation windows; censoring is separate.

Scope	CSR	Repair@2	@4	@8	Valid N
Whole object	—	—	—	—	—
Named module	—	—	—	—	—

Table S18. **E11: input modality and state assistance (planned).** RGB means images plus neutral rules and anonymous action handles. Text state can expose occluded geometry and is not information-equivalent to RGB. No unified text-state task list has been executed.

Input	Static SR	D0 BCR	CRR	Valid N
RGB + neutral task text	—	—	—	—
Explicit text state	—	—	—	—
RGB + explicit state	—	—	—	—

Table S19. **E12: output localization (planned).** Coordinates add pose estimation to action choice. Build the coordinate adapter and matched cases before comparison. Keep geometry, goals, and information fixed.

Output	SR/BCR	CRR	Schema errors	Valid N
Anonymous candidate ID	—	—	—	—
Free position + orientation	—	—	—	—

Table S20. **E13: independent snapshot detection (planned).** Includes normal controls and changed scenes. No in-loop diagnosis is required. Use multilabel location sets; the adapter and independent judgments are pending.

Agent	Precision	Recall	F1	False positive	Location F1
Human panel	—	—	—	—	—
Sol	—	—	—	—	—
Flash	—	—	—	—	—
Qwen 27B	—	—	—	—	—

Table S21. **E14: same-state cross-model repair (planned).** Select witnessed off-target states independently of recoverer outcomes. Every recoverer receives the identical state. Repeat by track; these restarts do not measure uninterrupted self-repair.

State source / recoverer	Sol	Flash	Qwen 27B
Sol	—	—	—
Flash	—	—	—
Qwen 27B	—	—	—

Table S22. **E15: observation robustness (unmeasured).** Keep camera/geometry fixed for resolution tests; restrict inspection too for single-view tests. Record delivered images and recheck visibility. Repeat separately by track and task type.

Condition	SR/BCR	CRR	Valid N
Native comparison canvas	—	—	—
One fixed view	—	—	—
No previous scene	—	—	—
Half image resolution	—	—	—
Double image resolution	—	—	—

Table S23. **E16: planning and computation (planned).** An open-loop script still executes one physical action at a time. Keep task lists and manipulation caps fixed; record computation and cost. No private reasoning trace is scored.

Strategy	SR/BCR	CRR	Decisions	USD
Standard closed loop	—	—	—	—
Open-loop action script	—	—	—	—
Explicit visual checking	—	—	—	—
Double decision allowance	—	—	—	—

Table S24. **E17: cost and action efficiency (unmeasured).** Include shared prefixes, both continuations, billed errors and retries. Attempt/reference is relative to a feasible witness, not proven optimality. Repeat per scope, task type, and model.

Model	Decisions	Inspect	Attempt/ref.	Image tokens	USD/task	USD/success
GPT-6.1 Sol	—	—	—	—	—	—
Claude Sonnet 5.5	—	—	—	—	—	—
Gemini 3.8 Flash	—	—	—	—	—	—
Claude Haiku 5.5	—	—	—	—	—	—
Qwen3.5-9B	—	—	—	—	—	—
Qwen3.5-27B	—	—	—	—	—	—
InternVL3.5-8B	—	—	—	—	—	—
Qwen2.5-VL-7B	—	—	—	—	—	—
Qwen2.5-VL-32B	—	—	—	—	—	—
Gemma 3-12B	—	—	—	—	—	—

Table S25. **E18: independent human operation (planned).** Graphical actions only. The proposed 72-condition panel spans static, whole-dynamic, and module-dynamic tasks; six participants provide three observations per condition without seeing both pair members. Current task selection and schedules require qualification. Independent humans are not cloned-prefix pairs; checkpoint comparisons are labeled C.

Agent	Static SR	BCR	ICR	TR	Valid N
Human (independent)	—	—	—	—	—
Sol	—	—	—	—	—
Flash	—	—	—	—	—
Qwen 27B	—	—	—	—	—

Table S26. **E19: validity and failure ledger (unmeasured).** Format, schema/IDs, physics, construction, exposure, and service outcomes are separate. Categories can co-occur; do not force their sum to equal failure N.

Record	Count	Rate / coverage
Planned / valid attempts	—	—
Parsing errors	—	—
Schema / handle errors	—	—
Physical rejections	—	—
Normal controls failed	—	—
Events untriggered	—	—
Events partially triggered	—	—
Incident effect unverified	—	—
Eligible CRR pairs	—	—
Model budget exhausted	—	—
Final service error	—	—
Harness error / external stop	—	—

M. Enumerated Configuration Inventories

The following tables enumerate all 63 primary dynamic configurations and all 160 original static source configurations. The dynamic inventory uses distinct current tasks and excludes diagnostic-only structures. The static inventory uses 3,265 distinct S tasks after 27 aliases; it is not the larger 3,697-task all-scope static main total. Parts refer to the active task or stage, never the entire supplied building.

Table S27. **All main dynamic operation configurations (1/2).** W: whole object; M: named remaining module. Counts are distinct dynamic tasks. Sibling static tasks and diagnostic structures are excluded.

Configuration	Scope	Parts	D0	D1	D2	Total
obj-airplane	W	8	1	38	0	39
obj-ambulance	W	10	1	24	0	25
obj-bed	W	9	1	38	0	39
obj-bench	W	11	1	33	0	34
obj-bookshelf	W	11	1	46	0	47
obj-bridge	W	15	1	27	0	28
obj-bunk-bed	W	11	1	22	0	23
obj-bus	W	8	1	29	0	30
obj-cabinet	W	8	1	46	0	47
obj-cargo-truck	W	10	1	41	0	42
obj-cash-register	W	6	1	22	0	23
obj-chair	W	7	1	20	0	21
obj-city-car	W	6	1	20	0	21
obj-clock-tower	W	13	1	45	0	46
obj-coffee-machine	W	9	1	28	0	29
obj-crane	W	9	1	22	0	23
obj-desk	W	13	1	40	0	41
obj-fire-engine	W	10	1	35	0	36
obj-forklift	W	15	1	38	0	39
obj-fountain	W	8	1	31	0	32
obj-gateway	W	15	1	35	0	36
obj-helicopter	W	6	1	16	0	17
obj-house	W	11	1	40	0	41
obj-lamp	W	7	1	13	0	14
obj-locomotive	W	11	1	37	0	38
obj-monitor	W	7	1	19	0	20
obj-motorboat	W	10	1	28	0	29
obj-piano	W	10	1	40	0	41
obj-pickup	W	9	1	38	0	39
obj-press	W	10	1	44	0	45
obj-rail-wagon	W	9	1	36	0	37
obj-refrigerator	W	14	1	45	0	46

Table S28. **All main dynamic operation configurations (2/2).** W: whole object; M: named remaining module. Counts are distinct dynamic tasks. Sibling static tasks and diagnostic structures are excluded.

Configuration	Scope	Parts	D0	D1	D2	Total
obj-robot	W	10	1	37	0	38
obj-rocket	W	12	1	56	0	57
obj-sofa	W	7	1	21	0	22
obj-staircase	W	11	1	39	0	40
obj-stool	W	6	1	17	0	18
obj-streetlight	W	7	1	12	0	13
obj-table	W	9	1	31	0	32
obj-tractor	W	11	1	34	0	35
obj-traffic-light	W	8	1	22	0	23
obj-washing-machine	W	10	1	49	0	50
cottage-roof	M	6	1	15	0	16
racer-top	M	11	1	16	3	20
seaplane-tail	M	7	1	13	2	16
museum-roof	M	12	1	14	2	17
repair-truck-cab	M	5	1	8	0	9
orange-helicopter-tail	M	7	1	12	2	15
police-helicopter-top	M	10	1	19	2	22
panda-head	M	6	1	12	0	13
burger-stand-roof	M	7	1	16	3	20
white-house-roof	M	10	1	17	3	21
rockefeller-crown	M	8	1	11	2	14
brandenburg-top	M	10	1	14	2	17
v7-long-bookshelf	W	21	1	18	4	23
v7-long-city-bus	W	32	1	18	4	23
v7-long-stone-bridge	W	25	1	18	4	23
v7-long-gantry-press	W	24	1	20	4	25
v7-long-cottage	W	24	1	18	4	23
museum-roof-long	M	16	1	18	2	21
whole-4018	W	12	1	6	0	7
whole-4017	W	13	1	6	0	7
whole-4037	W	9	1	6	0	7

Table S29. **Static source configuration inventory (1/2).** P: stage target parts; Q: distinct tasks after within-contract deduplication. All 160 configurations remain; Q sums to 3,265 across both tables.

Index	Configuration	P	Q	Index	Configuration	P	Q
S001	10001	11	56	S041	v23-10233	11	23
S002	10001-deep	16	48	S042	v23-10241	11	18
S003	10014	11	33	S043	v23-10248	11	25
S004	10036	11	73	S044	v23-10258	11	18
S005	10036-alt1	11	28	S045	v23-10261	11	17
S006	10128	11	24	S046	v23-10262	11	22
S007	10156	11	64	S047	v23-10267	11	18
S008	10156-alt1	11	20	S048	v23-10271	11	21
S009	10159	11	23	S049	v23-1472	11	18
S010	10213	11	62	S050	v23-1484	11	16
S011	10214	11	21	S051	v23-1489	11	23
S012	10220	11	28	S052	v23-1490	11	16
S013	31009	11	18	S053	v23-1490-alt1	11	27
S014	31027	11	22	S054	v23-1572	11	22
S015	31027-alt1	11	26	S055	v23-1572-alt1	11	23
S016	31027-kart	11	22	S056	v23-1632	11	19
S017	31028	11	6	S057	v23-1633	11	23
S018	31028-alt1	11	25	S058	v23-1682	11	16
S019	31028-sailboat	11	25	S059	v23-1775	11	14
S020	5867	11	22	S060	v23-1775-alt1	10	21
S021	omr-10242	11	16	S061	v23-1792	11	22
S022	omr-10265	11	9	S062	v23-1808	11	22
S023	omr-10269	11	3	S063	v23-1808-alt1	11	21
S024	omr-21309	11	19	S064	v23-1854	11	18
S025	omr-31031	11	20	S065	v23-1854-alt1	11	25
S026	omr-31088	11	6	S066	v23-20015	11	21
S027	omr-42004	11	18	S067	v23-20015-alt1	11	25
S028	omr-42020	11	18	S068	v23-21004	11	14
S029	omr-42020-alt1	11	17	S069	v23-21005	11	17
S030	omr-42054	11	18	S070	v23-21006	11	22
S031	omr-42061	11	18	S071	v23-21007	11	18
S032	omr-42066	11	7	S072	v23-21009	11	14
S033	omr-42102	11	17	S073	v23-21010	11	22
S034	omr-42102-alt1	11	9	S074	v23-21011	11	21
S035	v23-10037	11	15	S075	v23-21012	11	22
S036	v23-10124	11	22	S076	v23-21014	11	22
S037	v23-10158	11	29	S077	v23-21016	11	18
S038	v23-10184	11	17	S078	v23-21017	11	22
S039	v23-10194	11	17	S079	v23-21018	11	22
S040	v23-10226	11	24	S080	v23-21020	11	17

Table S30. **Static source configuration inventory (2/2).** P: stage target parts; Q: distinct tasks after within-contract deduplication. All 160 configurations remain; Q sums to 3,265 across both tables.

Index	Configuration	P	Q	Index	Configuration	P	Q
S081	v23-21021	11	22	S121	v23-31040	11	5
S082	v23-21022	11	22	S122	v23-31040-alt1	11	26
S083	v23-21024	11	22	S123	v23-31045	11	17
S084	v23-21029	11	22	S124	v23-31048	11	22
S085	v23-21030	11	22	S125	v23-31049	11	7
S086	v23-21037	11	22	S126	v23-31063	11	23
S087	v23-21045	11	22	S127	v23-31066	11	23
S088	v23-21302	11	17	S128	v23-31070	11	8
S089	v23-21303	11	21	S129	v23-31072	11	22
S090	v23-21318	11	22	S130	v23-31072-alt1	11	22
S091	v23-21319	11	23	S131	v23-31084	11	22
S092	v23-21321	11	17	S132	v23-31092	11	11
S093	v23-21325	11	21	S133	v23-31092-alt1	11	22
S094	v23-30022	7	1	S134	v23-3178	11	18
S095	v23-30023	7	20	S135	v23-3178-alt1	11	22
S096	v23-30025	11	11	S136	v23-3179	11	5
S097	v23-30025-alt1	11	20	S137	v23-3179-alt1	11	26
S098	v23-30026	11	21	S138	v23-3181	11	18
S099	v23-30026-alt1	11	20	S139	v23-3221	11	23
S100	v23-30027	11	8	S140	v23-3223	11	22
S101	v23-30027-alt1	11	6	S141	v23-3223-alt1	8	18
S102	v23-30029	11	19	S142	v23-3223-deep	16	27
S103	v23-30029-alt1	11	22	S143	v23-4018	9	21
S104	v23-30181	11	22	S144	v23-6678	11	22
S105	v23-30181-alt1	11	7	S145	v23-6678-deep	16	23
S106	v23-30285	11	3	S146	v23-6679	11	16
S107	v23-30311	11	22	S147	v23-6682	11	22
S108	v23-30311-alt1	11	24	S148	v23-6682-alt1	11	21
S109	v23-30312	11	20	S149	v23-6683	11	22
S110	v23-30472	11	15	S150	v23-6683-alt1	8	18
S111	v23-30472-alt1	11	24	S151	v23-6689	11	23
S112	v23-30542	11	12	S152	v23-6689-alt1	11	25
S113	v23-30542-alt1	11	20	S153	v23-6690	11	22
S114	v23-30571	11	22	S154	v23-6690-alt1	11	8
S115	v23-30571-alt1	11	25	S155	v23-6698	11	10
S116	v23-30571-deep	16	16	S156	v23-6698-alt1	11	24
S117	v23-31025	11	17	S157	v23-6699	11	17
S118	v23-31037	11	18	S158	v23-6699-alt1	11	17
S119	v23-31038	11	17	S159	v23-6745	11	22
S120	v23-31039	11	23	S160	v23-6753	11	23

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